MURAD BIN INAM TARAR

MECHATRONICS ENGINEER | ROBOTICS ENGINEER | CONTROL & AUTOMATION ENGINEER

Amsterdam, Netherlands | muradbininamtarar@gmail.com | +31 6 85003800 | linkedin.com/in/murad-bin-inam-tarar

PROFESSIONAL SUMMARY

Mechatronics Engineer with 7+ years in robotics, industrial automation and embedded systems, and 5+ years managing multidisciplinary teams from concept through to commissioning. Proven record delivering complex electromechanical, intralogistics and automation projects across defence, logistics and manufacturing. Hands-on in Siemens TIA Portal, S7-1200/S7-1500, WinCC SCADA, ROS2, industrial robots, machine design and system integration. Comfortable in client-facing environments, risk management and mentoring engineers to deliver on time and to quality. Based in Amsterdam with Dutch Residence Permit, immediately available and open to relocation.

AREAS OF EXPERTISE

✓ Siemens TIA Portal	✓ ROS / ROS2	✓ PLC Programming
✓ IIoT / Industry 4.0	✓ Embedded Systems	✓ SCADA / WinCC
✓ System Integration	✓ Machine Design	✓ Commissioning

CAREER HIGHLIGHTS

PLC Commissioning: Siemens TIA Portal PLC systems delivered for 3 logistics clients, full lifecycle FAT/SAT, £50K+ projects.
Engineering Leadership: 120-person engineering unit, 350+ systems, uptime +35%, failure rates -20%.
AI & Embedded Systems: Edge AI access control (Raspberry Pi, TensorFlow, OpenCV), 98.9% accuracy, 90% reduction in incidents.
Autonomous UAV: Autonomous quadcopter built from scratch, PID/LQR control, OpenCV vision, 23-min endurance.

PROFESSIONAL EXPERIENCE

Professional Development and Career Transition, Self-employed, *Amsterdam, Netherlands* | Feb 2026 – Present

Completing certifications and freelance automation consultancy while actively seeking engineering roles in the Netherlands.

- **Certifications:** Completing Siemens TIA Portal V20, Industrial Automation and PLC Specialization (Packt/Coursera) and Ignition SCADA Core Credential (Inductive University).
- **Visa and Credentials:** Relocated to the Netherlands on Orientation Year Visa (Zoekjaar), completed Nuffic/IDW degree equivalency evaluation for both MSc and BEng qualifications.

Robotics & AI Trainer, SIRM College, *London, UK* | Jul 2025 – Jan 2026

Robotics and automation training for engineering cohorts covering embedded systems, ROS and autonomous platforms.

- **Training Design:** Designed and delivered structured robotics and AI training programmes building capability across microcontrollers, embedded systems, ROS and autonomous platforms.
- **Mentoring:** Mentored 20+ engineering students through hands-on project work, translating system integration and control theory concepts into practical applied skills.

Technical Project Lead, Loyd International, *Pakistan / Remote (UK)* | Apr 2025 – Nov 2025

Concept-to-commissioning delivery of Siemens PLC-based automation systems for intralogistics clients.

- **PLC Commissioning:** Architected and delivered Siemens TIA Portal-based control systems for conveyor and intralogistics material handling applications, managing full project lifecycle from requirements through FAT/SAT to client commissioning on projects valued at £50K+.
- **IIoT Integration:** Integrated IIoT platforms with cloud CRM systems using MQTT and Azure IoT Hub, providing real-time operational visibility and reducing manual client reporting by 40%.
- **Team Leadership:** Directed a distributed multidisciplinary engineering team across software, controls and mechanical disciplines using Agile/Scrum and JIRA, maintaining delivery quality and collaboration standards.
- **Risk Management:** Governed technical risk across concurrent workstreams, conducting structured risk assessments and architecture reviews to ensure quality gates were met before client handover.

Automation & Process Optimisation Lead, The Co-operative, *London, UK* | Jul 2023 – Jan 2025

Process automation and continuous improvement leadership across inventory and logistics systems in a high-volume retail operation.

- **Automation Delivery:** Led a 12-member cross-functional team through concept-to-commissioning delivery of a cloud-integrated inventory automation system, achieving real-time stock visibility and predictive restocking.
- **Process Improvement:** Drove continuous improvement initiatives reducing operational waste by 28%, process efficiency gains of 15% and customer satisfaction improvement of 17% through systematic workflow automation.

- **Stakeholder Alignment:** Collaborated across departments to align technical upgrades with business KPIs, optimising team performance and technology ROI.

Engineering Lead (Automation & Systems), Defence Engineering Organization, *Pakistan* | Nov 2019 – Aug 2021

Leadership of a 120-person engineering unit responsible for maintenance, repair and commissioning of electromechanical systems.

- **Unit Leadership:** Led a 120-person multidisciplinary engineering unit responsible for repair and maintenance of 350+ military vehicles and complex electromechanical systems.
- **Facility Commissioning:** Architected and commissioned a full armoured vehicle repair facility integrating overhead lifting, pneumatic tooling, electromechanical workbenches and embedded diagnostics, cutting maintenance downtime by 25%.
- **Procurement:** Managed £250K+ procurement across 12+ suppliers, achieving 11% cost savings through structured vendor negotiation and risk management.
- **Reliability Engineering:** Improved system uptime by 35% and reduced equipment failure rates by 20% through structured root-cause analysis and iterative design refinement.

Mechatronics Engineer, Defence Engineering Organization, *Pakistan* | Nov 2017 – Nov 2019

Mechatronic system design, embedded control and AI-enabled solutions for defence applications across a 350-vehicle fleet.

- **Weapon System Automation:** Led machine design and system integration for weapon system automation, engineering a real-time embedded control and sensor feedback architecture that improved targeting accuracy by 45%.
- **Fleet Management:** Engineered an IoT-based fleet management platform with CMMS integration, eliminating 70% of unplanned downtime across a 350-vehicle fleet.
- **AI Access Control:** Built an AI-enabled access control system using Raspberry Pi, TensorFlow and OpenCV, achieving 98.9% recognition accuracy and a 90% reduction in unauthorised access incidents.

Research Intern (Robotics & Embedded Systems), NUST Robotics Lab, *Pakistan* | Jun 2016 – Jan 2017

Robotics research on multi-robot platforms developing control algorithms and trajectory planning for autonomous systems.

- **Industrial Robotics:** Programmed and integrated ABB industrial robots on multi-robot R&D platforms, developing control algorithms in MATLAB, Simulink and Embedded C for trajectory planning and system integration.
- **R&D Methodology:** Applied Agile prototyping methodologies to shorten R&D cycles by 20%, contributing to peer-reviewed academic publications on autonomous robotics systems.

Education

MSc Advanced Robotics (Merit), Queen Mary University of London, *London, UK* | Sep 2022 – Jan 2024

Relevant modules: Robotic Systems, AI/ML, Computer Vision, Sensors & Signal Processing, Digital Control

BEng Mechatronics Engineering (Medal of Excellence, High Achiever Award), National University of Sciences and Technology, *Pakistan* | Sep 2013 – Sep 2017

Relevant coursework: Control Systems, Mechatronics Design, Robotics, Embedded Systems, Machine Design

Additional Training & Certifications

Six Sigma Foundations, LinkedIn Learning | 2023

Siemens TIA Portal V20, Siemens TIA University/ Udemy | In progress

Industrial Automation & PLC Specialization, Packt / Coursera | In progress

Ignition SCADA Core Credential, Inductive University | In progress

Key Projects

5-DOF Mechatronic Robotic Arm: Full hardware-firmware-software stack, SolidWorks mechanical design, 3D-printed PLA structure, embedded C++ control with Bluetooth app, SD card automation and manual joint control modes.

ROS2 Motion Planning, Panda Robot Arm: Cubic trajectory generation and Cartesian path planning using MoveIt2, with real-time joint visualisation in RViz and rqt_plot. Python, ROS2, Linux.

Autonomous Animal Tracking Drone: Quadcopter built from scratch, APM 2.6, IMU, brushless motors, Android/OpenCV colour-tracking vision system, PID/LQR flight stabilisation, 23-minute endurance after weight-to-thrust optimisation.

Languages

English (Fluent) | Urdu (Native) | Dutch (A1, actively learning)

Technical Proficiencies

Siemens TIA Portal | S7-1200 / S7-1500 | WinCC | SCADA | HMI | PROFINET | MODBUS TCP/IP | OPC UA | IEC 61131-3 | Process Control | ROS / ROS2 | MoveIt | Motion Planning | Trajectory Planning | Robot Kinematics | ABB Industrial Robots | UAV Systems | Autonomous Systems | C | C++ | Embedded C | Python | RTOS | Arduino | Raspberry Pi | TensorFlow | OpenCV | IoT / IIoT | Azure IoT Hub | CMMS | SolidWorks | AutoCAD | MATLAB / Simulink | Simscape | Proteus | Git | Agile / Scrum | JIRA | Risk Management